NORMALIZED
MODAL
DISPLACEMENT
[–]

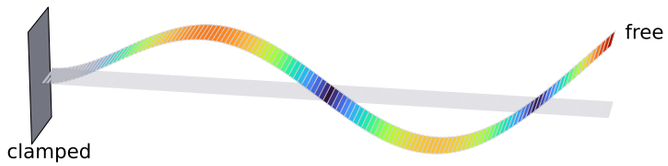
1.0

0.5

0.0

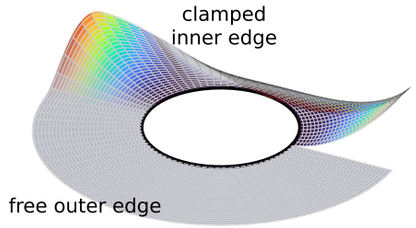
(a) P1 — clamped–free Euler–Bernoulli beam

analytic reference, modes 1–10 (shown: mode 3, $\Lambda = (\beta L)^4$)



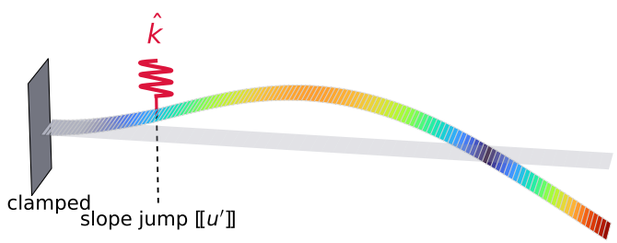
(b) P2 — annular Kirchhoff plate (reference only)

clamped inner / free outer, $a/b = 0.42$, $\nu = 0.29$; degenerate $\cos m\theta / \sin m\theta$ pairs (shown: $m = 2$, $n = 0$)



(c) P3 — beam with zero-width rotational spring

$x_c/L = 0.2$, $\hat{k} = k_\theta L/EI \in \{1, 10, 100, 1000\}$; $[\![u]\!] = 0$, $[\![u']\!]$ free (shown: $\hat{k} = 1$, mode 2)



(d) P4 — L-shaped plane-stress eigenproblem

unit arms, $\nu = 0.29$, clamped boundary, $\Omega^2 = \omega^2 \rho L^2/E$ (shown: mode 1)

